

Data Communication

Content 06: Switching

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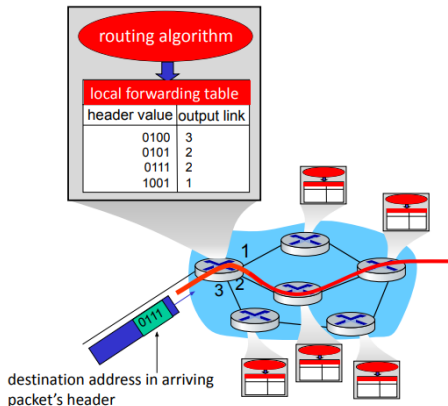
Switching

- **Switching** is the method of forwarding data within a network by selecting the optimal path through intermediate devices. Switching operates at the **Data Link Layer (Layer 2)**.
- **Routing** is the process of determining the best path to send data between different networks. Routing functions at the **Network Layer (Layer 3)** of the OSI model.
- Switching uses MAC addresses within a network; routing uses IP addresses to connect multiple networks.

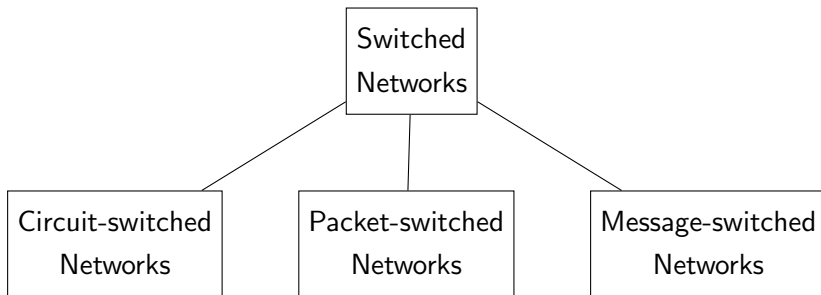


Switching & Routing

- Routing is a **global** action that determines source-destination paths taken by packets.
- Switching is a **local** action that moves arriving packets from router's input link to appropriate output link.



Switched Networks Classification

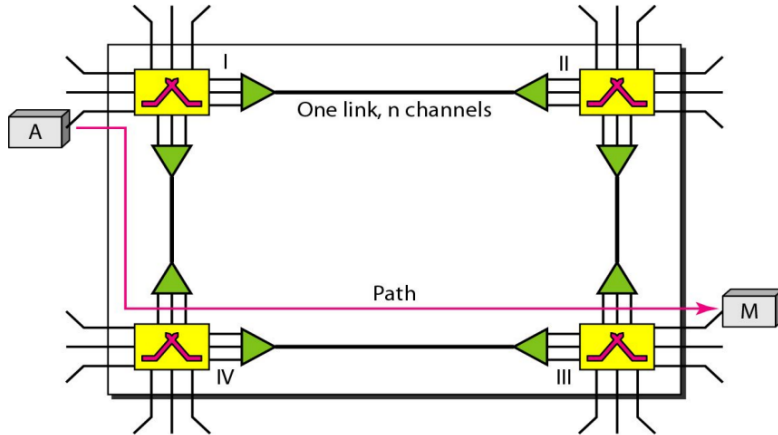


Circuit Switched Networks

- A **circuit-switched network** connects devices using switches and physical links.
- A **dedicated path** is created for each communication, and it stays reserved the whole time.
- Each link can carry multiple calls at once by dividing it into channels using **FDM** (Frequency Division Multiplexing) or **TDM** (Time Division Multiplexing).



Circuit Switched Networks



Circuit Switched Networks: Three Phase

In a circuit-switched network, a source and destination need to go through three phases. When end system A needs to communicate with system M:

1. **Setup Phase:** System A needs to request a connection to M that must be accepted by all switches as well as by M itself.
 - A channel is reserved on each link, and the combination of channels defines the dedicated path.
2. **Data Transfer:** After the dedicated path made of connected channels is established, data transfer can take place.
3. **Teardown Phase:** After all data have been transferred, the circuits are torn down.



Circuit Switched Networks: Efficiency & Delay

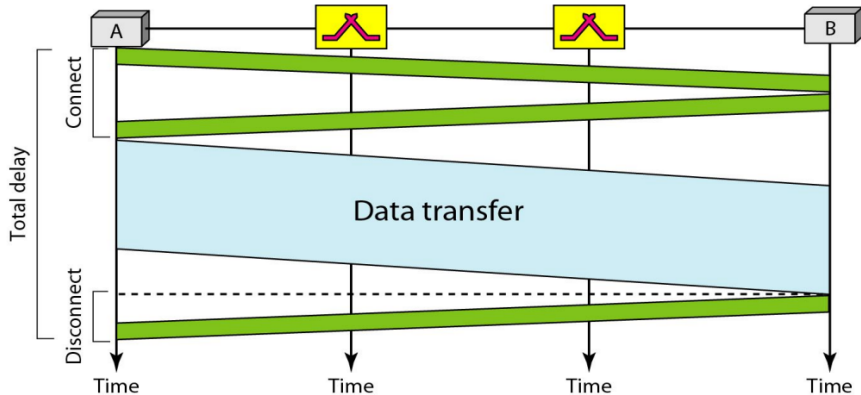
In circuit-switched networks, resources are reserved for the entire duration of the connection.

Example: Traditional telephone network.

- **Efficiency:** Low efficiency, as reserved resources remain unused by others even during silence.
- **Delay:** Minimal delay after setup, since a dedicated path is established.

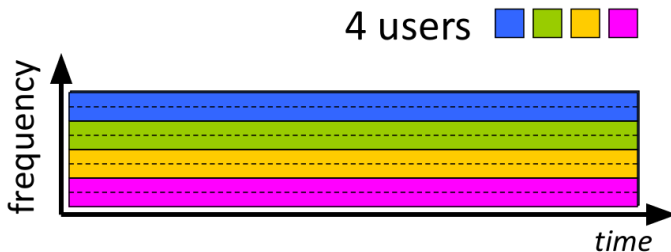


Circuit Switched Networks: Efficiency & Delay



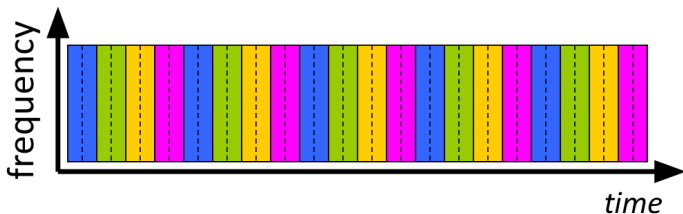
Circuit Switched Networks: FDM

- Optical, electromagnetic frequencies divided into (narrow) frequency bands
- Each call allocated its own band, can transmit at max rate of that narrow band



Circuit Switched Networks: TDM

- Time divided into slots
- Each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)

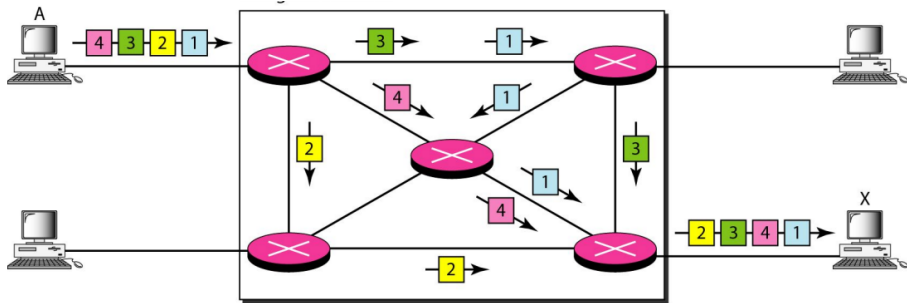


Packet Switched Networks

- In a **packet-switched network**, data is divided into small packets before being sent.
- Each packet is **routed independently**, possibly taking different paths to reach the destination.
- The network efficiently **shares resources** among many users without reserving a fixed path.



Packet Switched Networks



Packet Switched Networks: Efficiency & Delay

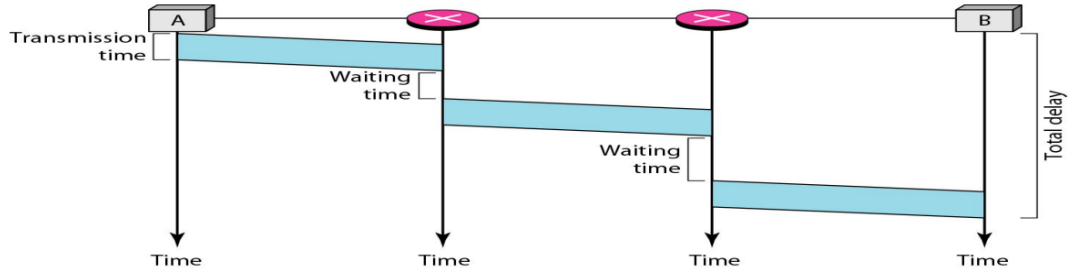
In packet-switched networks, data is divided into packets that are routed independently across the network.

Example: The Internet.

- **Efficiency:** High efficiency, as resources are shared among many users dynamically.
- **Delay:** Variable delay, since packets may take different paths and face congestion.

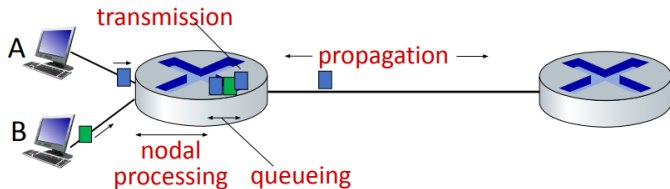


Packet Switched Networks: Efficiency & Delay



Delay

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

Delay

- **Propagation time:** The time required for a bit to travel from the source to the destination.

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

- **Transmission time:** The time between the first bit leaving the sender and the last bit arriving at the receiver.

$$\text{Transmission Time} = \frac{\text{Message Size}}{\text{Bandwidth (speed)}}$$



Delay

A sender wants to send a 2.5 KB file to a receiver 12,000 km away over a 1 Gbps network. What will be the propagation time and transmission time? Assume the speed of light in the medium is $2.4 * 10^8$ m/s.

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Propagation Speed}} = \frac{12000 * 10^3}{2.4 * 10^8} = 50 \text{ ms}$$

$$\text{Transmission Time} = \frac{\text{Message Size}}{\text{Bandwidth (speed)}} = \frac{2.5 * 10^3 * 8}{1 * 10^9} = 0.02 \text{ ms}$$



Circuit & Packet Switching: Pseudocode

Circuit Switching

```
setup_connection(source, destination)
route = routing_table_lookup(source, destination)
allocate_resources(route)

while data_available:
    transfer_data(route)

teardown_connection(route)
release_resources(route)
```

Packet Switching

```
for each packet in message:
    route = routing_table_lookup(
        current_router, packet.destination)
    forward_packet(packet, route)
```



Message Switched Networks

- A **message-switched network** stores the entire message at each switch before forwarding it — called *store-and-forward* transmission.
- No dedicated path is required, but each switch must have enough memory to hold the whole message.
- **Now obsolete** due to high delay, large storage needs, and the emergence of faster, more efficient **packet switching**.



References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 03 - 3.6
Chapter 08 - 8.1, 8.2
- Computer Networking: A Top Down Approach (6th Ed) - Jim Kurose
Chapter 01 - 1.3, 1.4



Thank You

